

Problem 2

Problem 2

Problem 1 Given the polynomial function q defined by $q(v) = 2v^3 - 5v^2 + 7v - 9$ find:

- (a) The slope of the tangent line to the graph of q at the point where $v = 3$ using the limit definition of a derivative.

Explanation. To compute $q'(3)$ from the limit definition we use either

$$\lim_{v \rightarrow 3} \frac{q(v) - q(3)}{v - 3} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{q(3 + h) - q(3)}{h}.$$

Using the second formula:

$$\begin{aligned} q'(3) &= \lim_{h \rightarrow 0} \frac{q(3 + h) - q(3)}{h} \text{ which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0} \frac{(2(3 + h)^3 - 5(3 + h)^2 + 7(3 + h) - 9) - (54 - 45 + 21 - 9)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2(27 + 27h + 9h^2 + h^3) - 5(9 + 6h + h^2) + 21 + 7h - 9) - 21}{h} \\ &= \lim_{h \rightarrow 0} \frac{54 + 54h + 18h^2 + 2h^3 - 45 - 30h - 5h^2 + 21 + 7h - 9 - 21}{h} \\ &= \lim_{h \rightarrow 0} \frac{31h + 13h^2 + 2h^3}{h} \\ &= \lim_{h \rightarrow 0} (31 + 13h + 2h^2) \\ &= 31. \end{aligned}$$

- (b) The slope of the tangent line to the graph of q at the point where $v = 3$ using the “short-cut derivative rules” to find a formula for q' and evaluating $q'(3)$.

Explanation.

$$\begin{aligned} q'(v) &= (2v^3 - 5v^2 + 7v - 9)' \\ &= (2v^3)' + (-5v^2)' + (7v)' + (-9)' \\ &= 3 \cdot 2 \cdot v^{3-1} + -5 \cdot (2) \cdot v^{2-1} + 7 \cdot 1 \cdot v^{1-1} + 0 \\ &= 6v^2 - 10v + 7 \\ &\implies q'(3) = 6 \cdot 3^2 - 10 \cdot 3 + 7 = 31. \end{aligned}$$

Problem 2

- (c) *The equation of the tangent line to the graph of q at $v = 3$.*

Explanation. Slope of tangent line to the graph of q at the point where $v = 3$: $q'(3) = 6 \cdot 3^2 - 10 \cdot 3 + 7 = 31$.

Point on tangent line at the point where $v = 3$: $(3, q(3)) = (3, 21)$.

Equation of tangent line to the graph of q at the point where $v = 3$:

$$\begin{aligned} y - 21 &= 31(x - 3) \\ \implies y &= 31x - 72. \end{aligned}$$